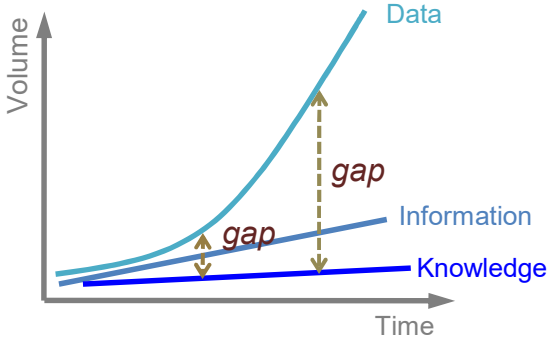
Computational Challenges in HTS-based bioinformatics



- 数据规模庞大
- 数据类型复杂
- 技术门槛偏高
- 可重复性不强
- 方法学不成熟



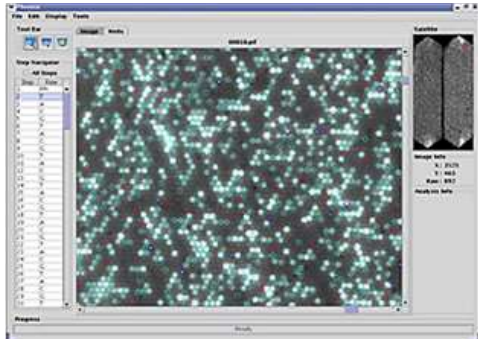


(i)
Sequencing bias/errors

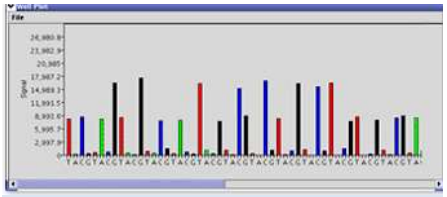


Sequencing bias/errors

- **454**



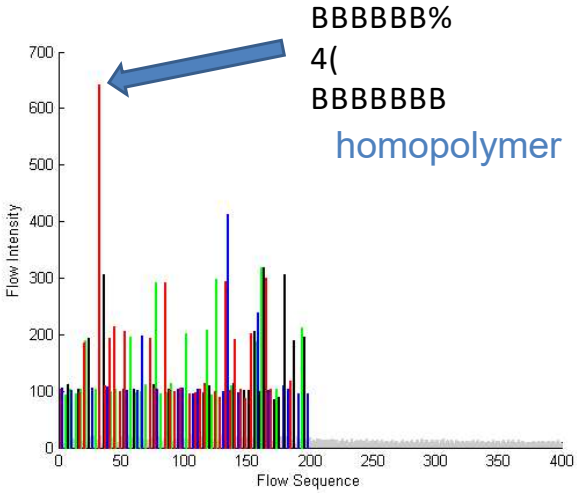
The light is detected by a CCD camera, its intensity proportional to the number of sequential bases that were added



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!"#$$!$$"#!!$""=

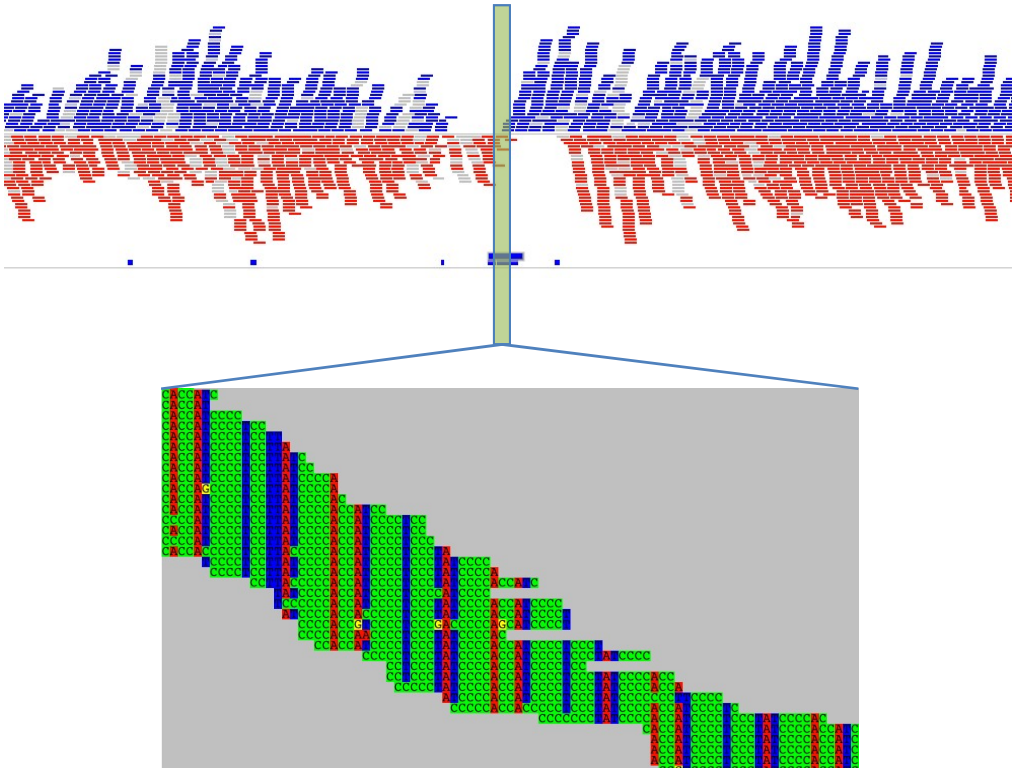
This results in a series of peaks that follows the order of added nucleotides and may be read to create a sequence



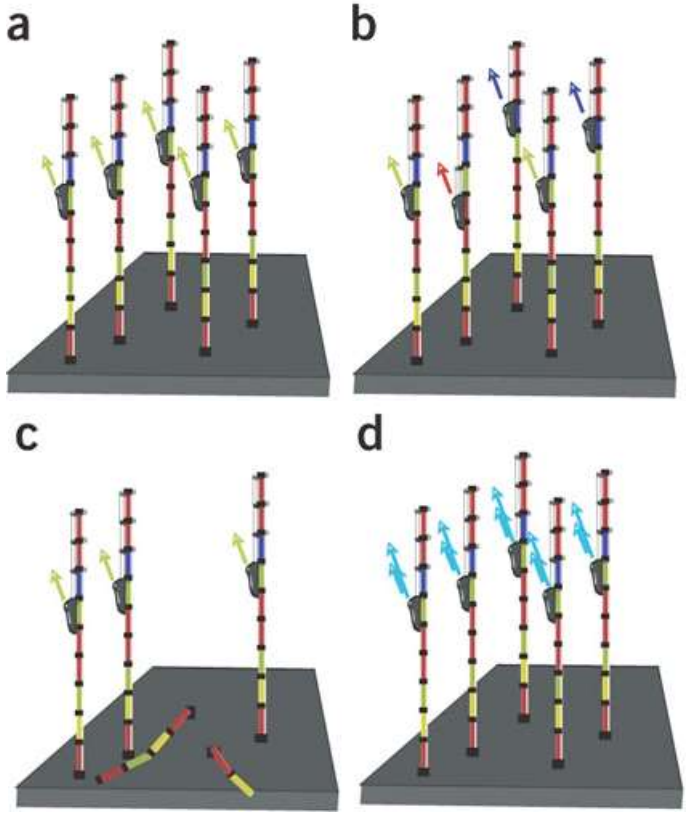
When the same base appears several times in a row, it may be hard to distinguish the exact brightness of the light

Sequencing bias/errors

- **Illumina**



High GC

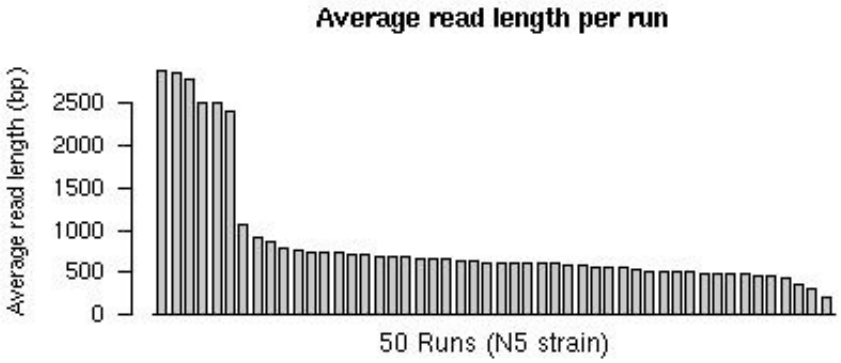


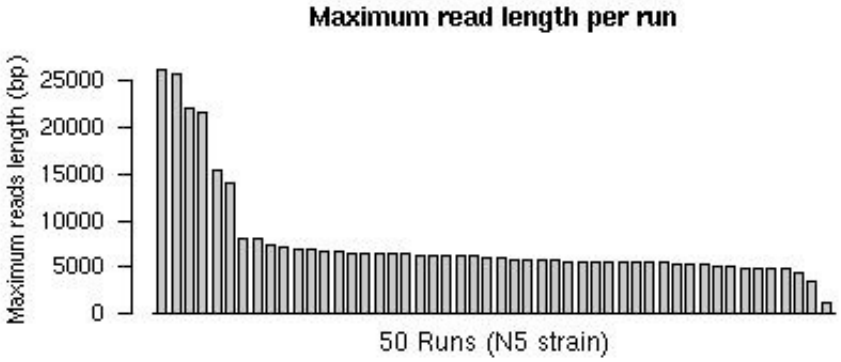
Accumulated errors arise from:
(b) Phasing noise
(c) Fading
(d) Fluorophore cross talk

(Erlich et al. Nature Methods 2008)

Sequencing bias/errors

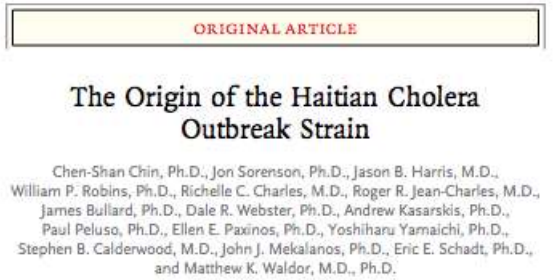
- **PacBio**





the total amount of raw sequence generated per run ranged from 6.6Mb to 136.3Mb. The combined amount for the 50 runs was 2.0Gb.

~25% of reads could be mapped
~20% error rate compared to ref


ORIGINAL ARTICLE

The Origin of the Haitian Cholera Outbreak Strain

Chen-Shan Chin, Ph.D., Jon Sorenson, Ph.D., Jason B. Harris, M.D., William P. Robins, Ph.D., Richelle C. Charles, M.D., Roger R. Jean-Charles, M.D., James Bullard, Ph.D., Dale R. Webster, Ph.D., Andrew Kasarskis, Ph.D., Paul Peluso, Ph.D., Ellen E. Paxinos, Ph.D., Yoshiharu Yamaichi, Ph.D., Stephen B. Calderwood, M.D., John J. Mekalanos, Ph.D., Eric E. Schadt, Ph.D., and Matthew K. Waldor, M.D., Ph.D.

N Engl J Med January 6, 2011

Possible solutions

1. Deep sequencing
2. Statistical evaluation
3. Error correction

Correcting Errors in Short Reads by Multiple Alignments

Leena Salmela[1]* and Jan Schröder[2]

[1]Department of Computer Science and Helsinki Institute for Information Technology HIIT, University of Helsinki, Finland
[2]NICTA Victorian Research Lab, Department of Computer Science and Software Engineering, University of Melbourne, Australia

Kelley et al. Genome Biology 2010, 11:R116
http://genomebiology.com/2010/11/11/R116

Genome Biology

SOFTWARE Open Access

Quake: quality-aware detection and correction of sequencing errors

David R Kelley[1*], Michael C Schatz[2], Steven L Salzberg[1]

Resource

ECHO: A reference-free short-read error correction algorithm

Wei-Chun Kao,[1] Andrew H. Chan,[1] and Yun S. Song[1,2,3]

[1]Computer Science Division, University of California, Berkeley, California 94721, USA; [2]Department of Statistics, University of California, Berkeley, California 94721, USA

Bartonella Henselae
330X coverage, 36bp Illumina reads

	Avg Contig	Max Contig	N50	N75	Contigs
Before	4,474	29,042	8,260	3,144	374
After	7,094	79,281	17,397	4,675	231